

Operating Manual

Translation of the Original Operating Manual

Wire Straightener 2700



Schäfer Werkzeug- und
Sondermaschinenbau GmbH

Dr.- Alfred -Weckesser-Straße 6
76669 Bad Schönborn

Tel: 07253/9421-0
Fax: 07253/9421-94

www.schaefer.biz
info@schaefer.biz

Edition 2016-07-19

This document is subject to change without notice

Safety Rules

 DANGER!	DANGER indicates a hazardous situation which, if not avoided will result in death or serious injury.
 WARNING!	WARNING indicates a hazardous situation which, if not avoided could result in death or serious injury.
 CAUTION!	CAUTION indicates a hazardous situation which, if not avoided may result in minor or moderate injury.
 NOTICE	NOTICE is used to address practices not related to physical injury.
 INFO	Additional Information

Table of Contents:

1	Introduction and Things to Know	5
1.1	Introduction	5
1.2	Intended Use	5
1.3	Nameplate	5
1.4	Technical Data	6
2	Safety	6
2.1	Safety-relevant Components	6
2.2	Protective guard:	6
2.3	General Safety Rules	6
3	Receiving Inspection	7
4	Setting the Machine Up	7
4.1	Operator's Workplace	8
4.2	Electrical/Pneumatic Connections	8
4.3	Attachment Parts	9
4.3.1	Protective Guards	9
4.3.2	Pressure Adjusting Knob	9
5	Control and Monitoring Elements	10
5.1	MAIN Switch	10
5.2	± Pushbuttons for the Cycle Time	10
5.3	Pressure Adjusting Knob	11
6	Setup and Maintenance Work	11
6.1	Adjustment Work	11
6.1.1	Setting the stroke parameter	11
6.2	Maintenance Work	11
6.2.1	Cleaning	11
7	Working with the Wire Straightener	12
8	Storage	12
9	Disassembly	12
10	Disposal	13
11	Fault Analysis	13
11.1	General Faults	13
12	Control System	13
13	Contact	14

1 Introduction and Things to Know

1.1 Introduction

Dear Customer:

Thank you for purchasing this modern, highly efficient and reliable wire straightener which helps you perform your daily production duties.

Prior to commissioning of the machine, it is recommended that the user thoroughly reads and understands **all chapters**. In this way unnecessary time losses are avoided and risks of operating errors ruled out.

In the event of questions, please feel free to contact us anytime.

Sincerely yours,

Schäfer GmbH

1.2 Intended Use

The duty of this wire processing machine is to straighten two-wire sheathed cables.

The machine must not be operated outdoors.

Using the plant in an hazardous environment is prohibited.

The machine must exclusively be operated in dry and clean rooms (low-dust environment).

Reconstruction or add-on measures are inadmissible unless the original accessories furnished by us are employed.

Schäfer GmbH will not assume any liability for damage caused by other or deviating uses or applications or by attachment parts that have not been approved by us.

1.3 Nameplate

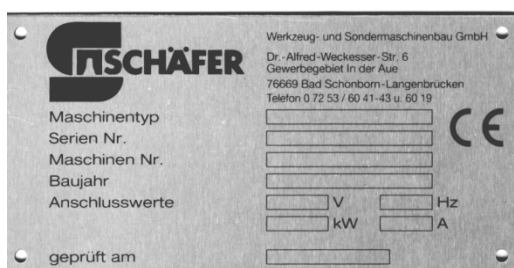


Figure 1

The nameplate has been mounted on the left side of rack and includes important data such as machine type, machine-number, year built and connected load data.

1.4 Technical Data

Dimensions:	
• Length	appr. 230 mm
• Width	appr. 300 mm
• Height	appr. 200 mm
Weight, complete:	appr. 10 kg
Connected electrical load:	110 V / 230V, 1-phase, 50 / 60 Hz, 0.25 A
Pneumatic system operating pressure:	6 bar
Cycle time	Adjustable

2 Safety

2.1 Safety-relevant Components

 WARNING!	Before connecting the machine to the power systems be sure to familiarize yourself with the control elements (Page 11, Chapter 5)!
--	--

Main switch:

During set-up, maintenance and repair work this switch must always be in position „0“ to rule out inadvertent switching which may cause injuries (e.g. of the fingers).

2.2 Protective guard:

 WARNING!	Never perform work with the protective guard removed! Strictly observe the special warning notes for operators: Do not remove warning notes. Replace them if illegible.
--	--

2.3 General Safety Rules

The Owner of the machine or a person authorized by him will be orally instructed by Schäfer GmbH how to control and run the plant.

It is within the Owner's scope of responsibility to appropriately familiarize the (authorized) set-up personnel as well as machine operators with procedures to be followed.

Before the machine is set up and commissioned all chapters shall be carefully studied and be understood.

 DANGER!	When machine has been opened service work must exclusively be performed with the mains cable disconnected and after a 3-minute waiting period has elapsed. Otherwise, danger to life is imminent due to electric shock!
 WARNING!	• Setup and maintenance work on the machine must only be carried out by trained personnel duly authorized and familiarized by the Plant Owner! • All set-up work on the machine must be performed with the system switched off (main switch in „0“ position) - otherwise, injuries may be caused due to inadvertent activation! • Make sure not to operate the machine in case of malfunctions or defective parts! • Observe all other relevant safety regulations!
 WARNING!	• Only use wire of correct cross sectional area. • Do not put tools or other items or objects on the machine. • Reconstruction of or other modifications to the machine are prohibited. • Exclusively use original attachment parts and accessories.
 NOTICE	• Always keep this operating manual ready at the machine's location. It must be available to operators and setup staff whenever needed. • The operating manual is to be viewed as integral part of the machine and must be handed over to buyer upon sale of the machine.

3 Receiving Inspection

Check the completeness of the consignment by way of the list of materials below. Please inform us without delay if parts are missing, and in the event of transit damage induce forwarder to confirm that such damage has occurred.

1. 1x Wire straightener
2. 2x Operating manual

4 Setting the Machine Up

 CAUTION!	The mounting location of the machine must be stable, level, dry and free of oil or grease!
--	---

4.1 Operator's Workplace

When setting up the equipment observe the following:

- The workplace of the operator is in front of the wire straightener.
- In the interest of fatigue-free working the workplace must be designed to satisfy ergonomic needs. This requires provision of chairs adjustable in height and pedestals for shorter persons.
- Make sure that there is a good illumination at the workplace. If necessary, install additional lights.

4.2 Electrical/Pneumatic Connections

 DANGER!	Only skilled electrical personnel must connect the equipment to the electrical power system!
 WARNING!	Before connecting the machine make sure that the main switch is in position „0“!
 CAUTION!	Prior to initiation, check whether voltage and air pressure are correctly set.

Electrical system:

For the standard design machine the mains connection is **110 V / 230 V, 50 / 60 Hz**.

Pneumatic system:

A pressure adjusting knob is arranged on the machine by means of which air can be treated and adjusted to the required pressure.

Connection is made via a coupling plug type KS-PK-4.

Compressed air to be used must be in accordance with DIN ISO 8573 and be of quality grade 3.3.5.

4.3 Attachment Parts

4.3.1 Protective Guards

 WARNING!	In the interest of safe working and minimizing injury hazards, it is absolutely necessary to mount the protective guards correctly! Exclusively use original attachment parts and accessories!
--	---

This machine is delivered with the below described protective guards already mounted. Owner is not permitted to modify or reconstruct the machine.

1. Protective Hood 2700.0016

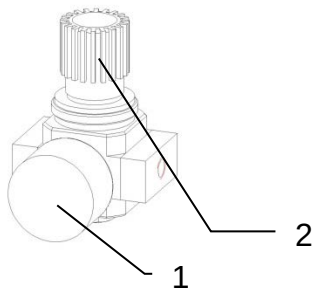
4.3.2 Pressure Adjusting Knob

The operating pressure is adjusted via knob **(2)** on top.

In direction of arrow + = pressure will increase

In direction of arrow - = pressure will decrease

Arrest the pressure adjusting knob by pressing it down.



- 1) Pressure gauge
- 2) Pressure adjusting knob

Figure 2

5 Control and Monitoring Elements

The control and monitoring elements of the wire straightener are located above the mechanical machine components.

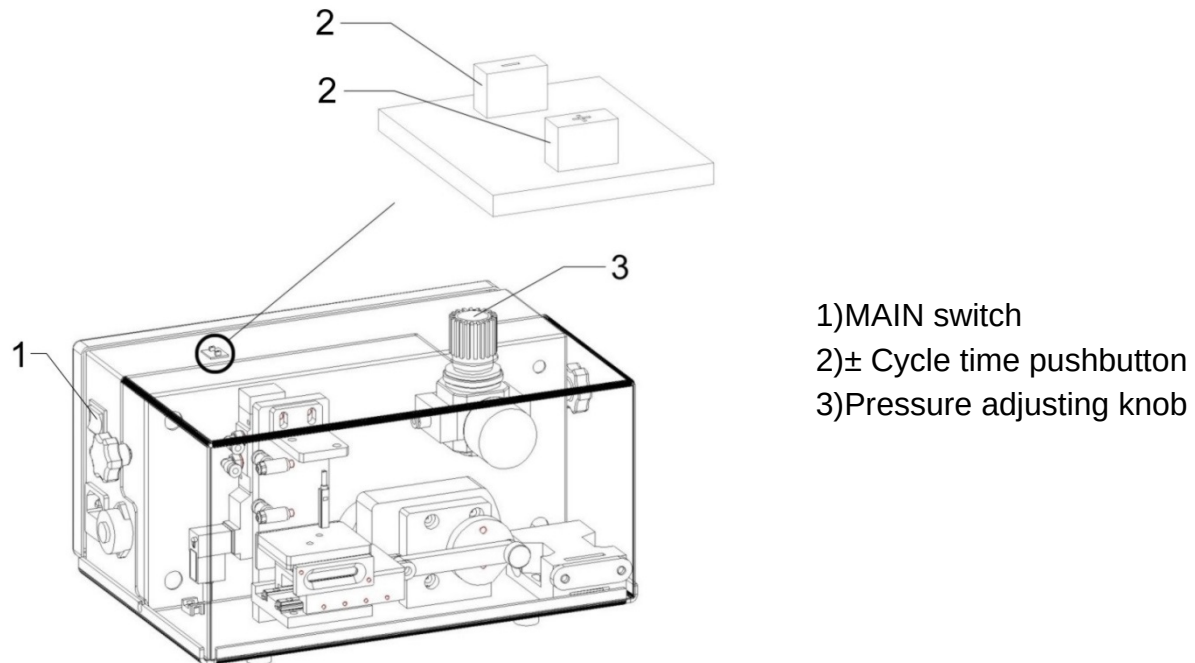


Figure 33

5.1 MAIN Switch

 WARNING!	Carry out set-up and service work only if the main switch is in position „0“!
--	--

Changing the main switch position from „0“ to „1“, connects the wire straightener to the power supply. The straightener is now ready to operate.

5.2 ± Pushbuttons for the Cycle Time

Use the ± buttons to adjust the cycle time as necessary. You can see the adjusted number of cycles (0-9) on the indicator between the buttons.

A cycle equates to 0.25 seconds.

Shortest time span, 0 shown on display = 0.5 sec.

+ = Increase cycle time

- = Reduce cycle time

5.3 Pressure Adjusting Knob

This knob serves to adjust the pressure between thrust plant and slide.

6 Setup and Maintenance Work

6.1 Adjustment Work

6.1.1 Setting the stroke parameter

The stroke can be adjusted as follows:

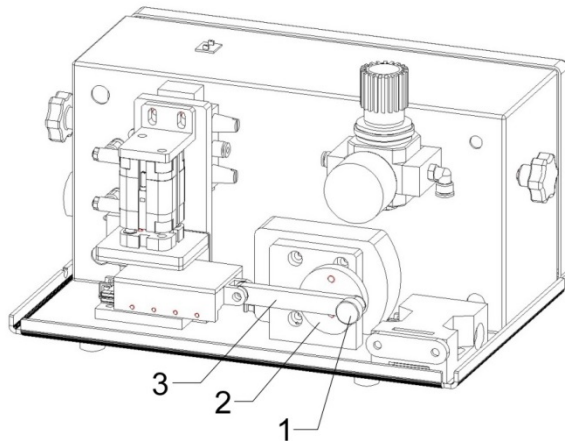


Figure 4

1. Remove the collar screw (1).
2. Place connecting rod (3) in the desired opening of eccentric disk (2) and secure by tightening the collar screw (1).

- A) Stroke length = 18 mm
- B) Stroke length = 24 mm
- C) Stroke length = 30 mm
- D) Stroke length = 36 mm

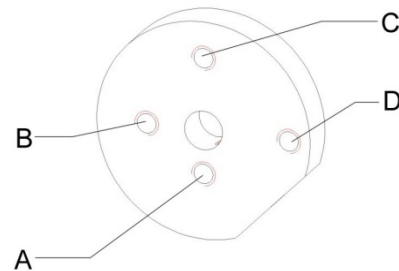


Figure 5

6.2 Maintenance Work

6.2.1 Cleaning

Remove coarse dirt at least once a day, preferably by means of a brush.

7 Working with the Wire Straightener

- MAIN switch in position „0“.
- If necessary, connect foot pedal and/or compressed air (**see chapter 4.2**).
- Plug in the mains cable.
- Change MAIN switch from „0“ to „1“ position.
- Insert the twisted wire through the feed opening.
- Actuate the foot pedal.
- Remove the wire

8 Storage

 CAUTION!	Keep the unit dry and protect it against humidity! The machine must not be stored outdoors!
--	--

In the event that the machine is to be shut down for a longer period:

- Unplug the mains cable
- Disconnect the machine from the compressed air system
- Clean the unit
- Grease the unit properly
- Apply a corrosion protective coat, especially to all bright parts (e.g. use fine spray agent type OKS 701 or similar)
- Store unit in a dry and clean location

9 Disassembly

- Disconnect the machine from the mains (unplug mains cable).
- Uncouple the pneumatic connection.

Dismantling of the machine by the customer to an extent greater than described in the operating manual is prohibited. Should such dismantling be required the machine must be returned to Schäfer GmbH.

10 Disposal

After the service life of the machine has come to an end its components must be disposed of or recycled expertly.

- Dismantle the unit to the extent permissible.

The individual components sorted according to relevant materials must then be processed by recycling and disposal specialist companies.

 NOTICE	Electronic components consist of a variety of problematic or hazardous materials. Therefore, the parts of the electronic system of the press must by no means be mixed up with the regular waste. There are specialist companies dealing with the recycling of electronic scrap material.
--	--

11 Fault Analysis

 DANGER!	The troubleshooting steps marked with an * (asterisk) must be carried out by electrical personnel only! Observe the safety rule: the main switch must be in position „0“!
---	---

11.1 General Faults

Symptoms	Remedial Action
Pilot lamps do not light up	• Plug in the mains cable • Mount the protective hood • Main switch must be in position „1“ • Check mains voltage and fuses*
Machine does not start	• Check compressed air supply

12 Control System

 DANGER!	Prior to opening the switchboard unplug the mains cable! Work on live electrical components must exclusively be performed by skilled expert personnel trained in electrical engineering!
---	--

13 Contact

Please contact directly one of the numerous service and sales locations representing the Schäfer Group. An updated list of addresses by means of which you may contact us worldwide is available on our website.

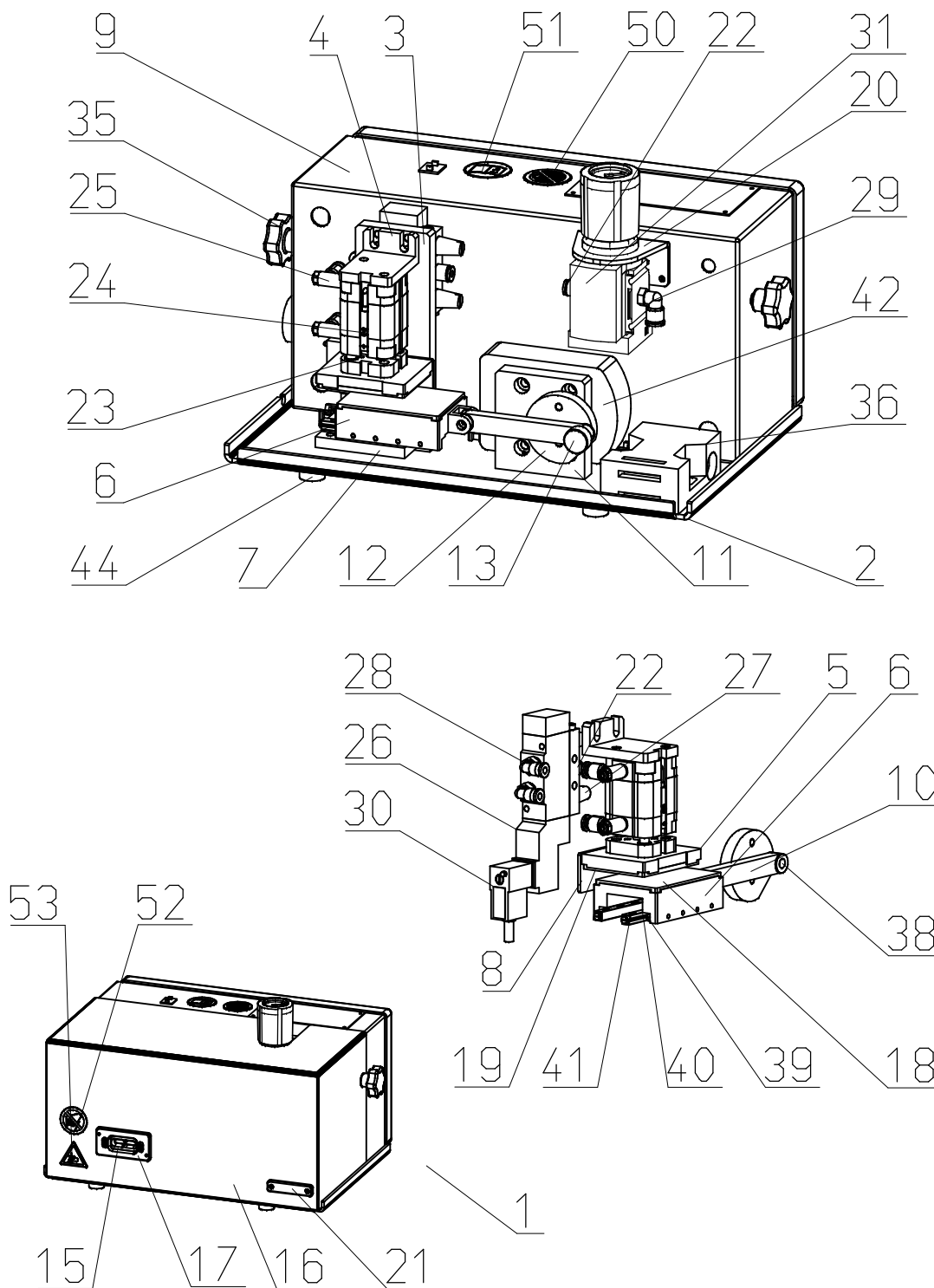


Schäfer Werkzeug- und
Sondermaschinenbau GmbH

Dr.- Alfred -Weckesser-Straße 6
76669 Bad Schönborn

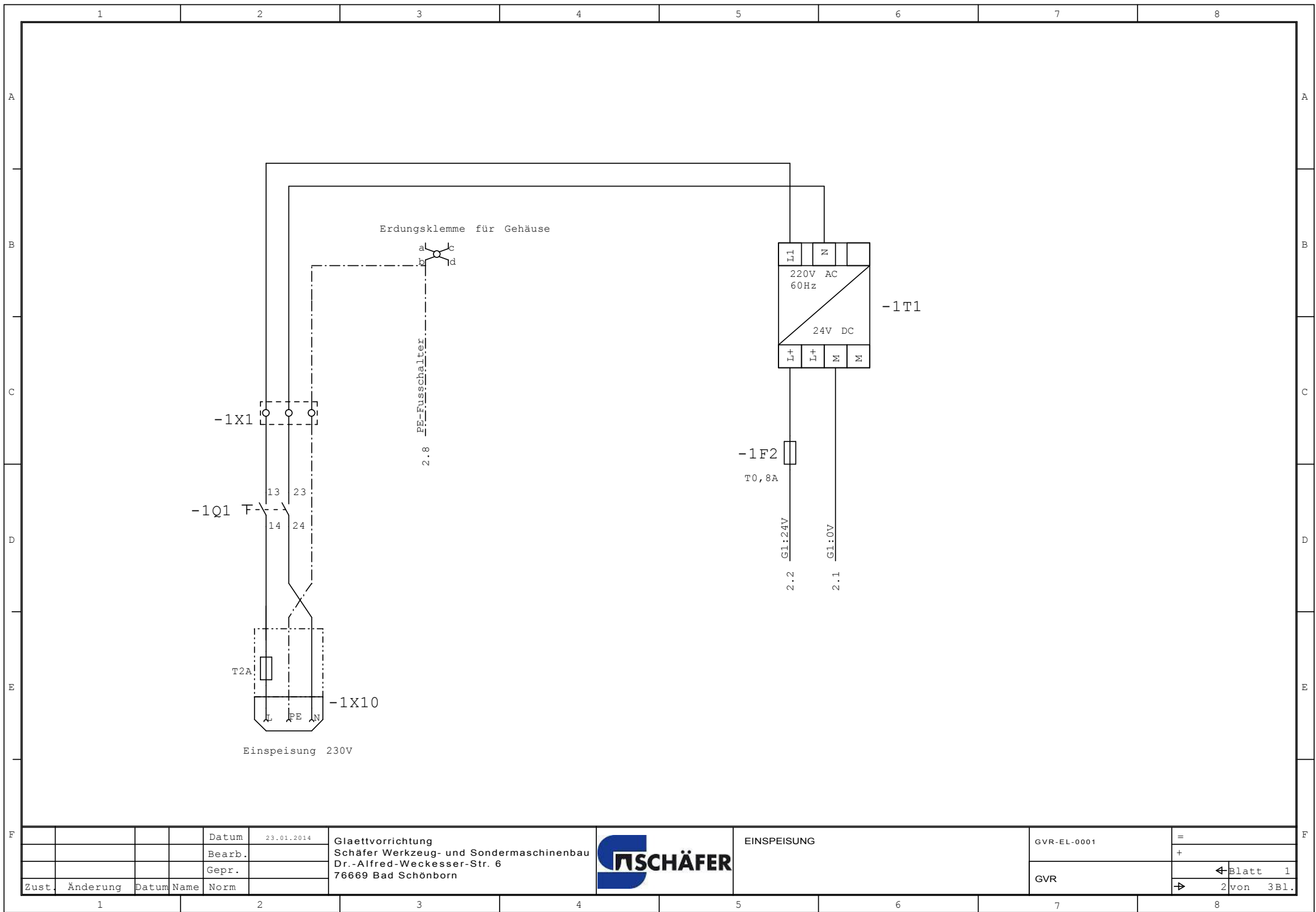
Tel: 07253/9421-0
Fax: 07253/9421-94

www.schaefer.biz
info@schaefer.biz



53	1	Warnzeichen-Warnung vor Handverlet...	10628323	10628323
52	1	Verbotszeichen-Hineinfassen verbot...	10628321	10628321
51	1	Gebotszeichen-Gebrauchsanweisung b...	10628322	10628322
50	1	Verbotszeichen-Schutzeinrichtung e...	10628318	10628318
44	4	Fuss Gummi Stehbolzen	10610266	10610266
43	3	Kabelbinder-Schelle M3	10608924	10608924
42	1	Getriebemotor	10616709	10616709
41	8	Endstueck GB2	10614036	10614036
40	4	Linearfuehrung	10614035	10614035
39	2	Rollenkaefig	10614667	10614667
38	4	PTFE Composite-Bundbuchse	10613236	10613236
37	1	gerader Betaetiger	10600972	10600972
36	1	Sicherheitsschalter	10600161	10600161
35	2	Sterngriff M5x10	10611549	10611549
31	1	Druckregelventil	10629198	10629198
30	1	Ventilanschlusskabel Bauform C/5m/...	10601934	10601934
29	1	L-Steckverschraubung, 1/8-6	10601252	10601252
28	2	L-Steckverschraubung QSML-1/8-4	10601247	10601247
27	2	Schalldaempfer, G1/8a	10602217	10602217
26	1	Magnetventil	10601180	10601180
25	2	Drosselrueckschlagventil, 4xM5	10612583	10612583
24	1	Naehierungsschalter	10626991	10626991
23	1	Kompaktzylinder	10616704	10616704
22	2	Steckverschraubung, QSM-1/8-4-I	10600334	10600334
21	1	Befestigungsplatte	10592736	2700.0016-1-4.0
20	1	Aufnahmewinkel	10592746	2700.0030-0001
19	1	Glaettplatte	10590529	2700.0021
18	1	Glaettplatte	10590528	2700.0020
17	1	Aufnahme Einfuehrplatte	10592734	2700.0015-1-0002
16	1	Schutzhaube	10592735	2700.0016-0002
15	1	Einfuehrplatte	10592731	2700.0015-0002
13	1	Bundschraube	10541178	2700.0009
12	1	Exzenterscheibe	10541177	2700.0008
11	1	Motorbefestigungsplatte	10541260	2700.0010
10	1	Pleuel	10541264	2700.0007
9	1	Nacharbeit Klemmkasten	10552500	2700.0011
8	1	Anschlagplatte	10549458	2700.0012
7	1	Fuehrungsplatte	10592465	2700.0003-0001
6	1	Schlitten	10592464	2700.0004-0002
5	1	Anpressplatte	10569628	2700.0006-0001
4	1	Zylinderbefestigung	10541262	2700.0005
3	1	Befestigungsplatte	10541256	2700.0002
2	1	Grundblech	10541137	2700.0001
1	1	Glaettvorrichtung	10400626	2700.0000.01-0003
Pos.	Anz.	Benennung	Identnr.	Artikelnummer

Stck.: 1	Kantenbearbeitung :			Fachnummer :	
Oberflaechenbeschaffenheit :	Allgemeintoleranz DIN 2768-		Blattformat DIN A3	Maßstab : 1:3	CAD-Zeichnung, alle Rechte vorbehalten, Schutzvermerk nach DIN ISO 16016
	Massangaben in mm				
Werkstoff :		Datum	Name	Zeichnungsnummer: 2700.0000.01-0003	
	Bearb.	10.03.2017	J.Conrad	Zusammenbau Glaettvorrichtung	
	Gepr.	04.04.2017	J.Conrad		
Waermebehandl ung :	Dateiname: C:00145731				
Oberflaechenbehandl ung:				Identnummer: 10400626	Ausgabedatum 04.04.2017



				Datum	23.01.2014
				Bearb.	
				Gepr.	
Zust.	Änderung	Datum	Name	Norm	

Glaettvorrichtung
Schäfer Werkzeug- und Sondermaschinenbau
Dr.-Alfred-Weckesser-Str. 6
76669 Bad Schönborn

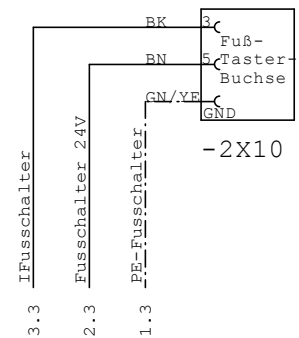
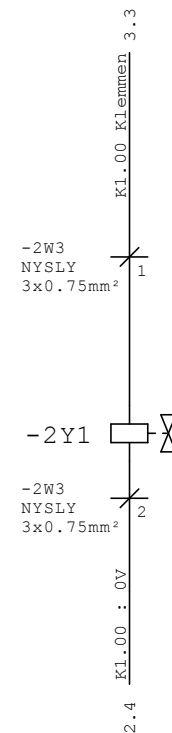
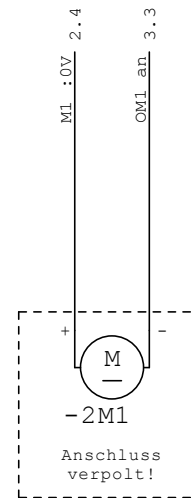
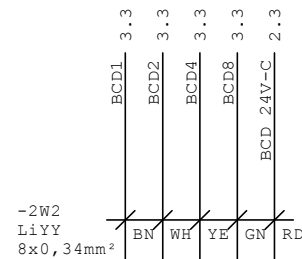
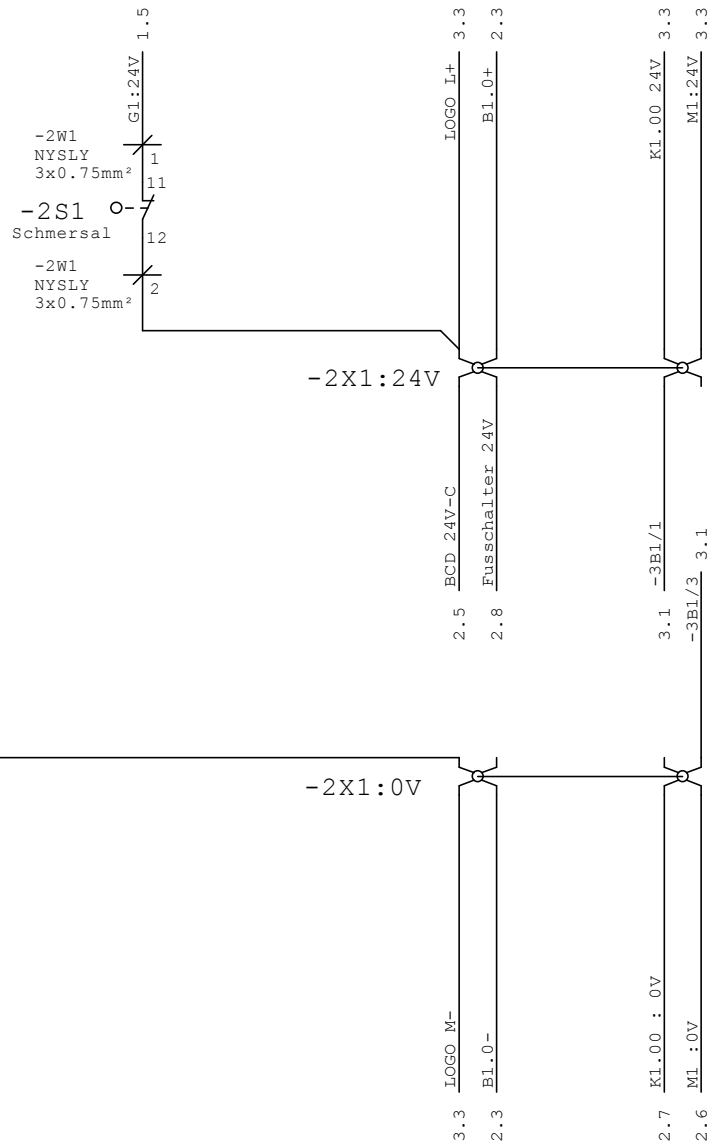


EINSPEISUNG

GVR-EL-0001

GVR

=
+
← Blatt 1
→ 2 von 3 Bl.



				Datum	23.01.2014
				Bearb.	
				Gepr.	
Zust.	Änderung	Datum	Name	Norm	

Glaettvorrichtung
Schäfer Werkzeug- und Sondermaschinenbau
Dr.-Alfred-Weckesser-Str. 6
76669 Bad Schönborn

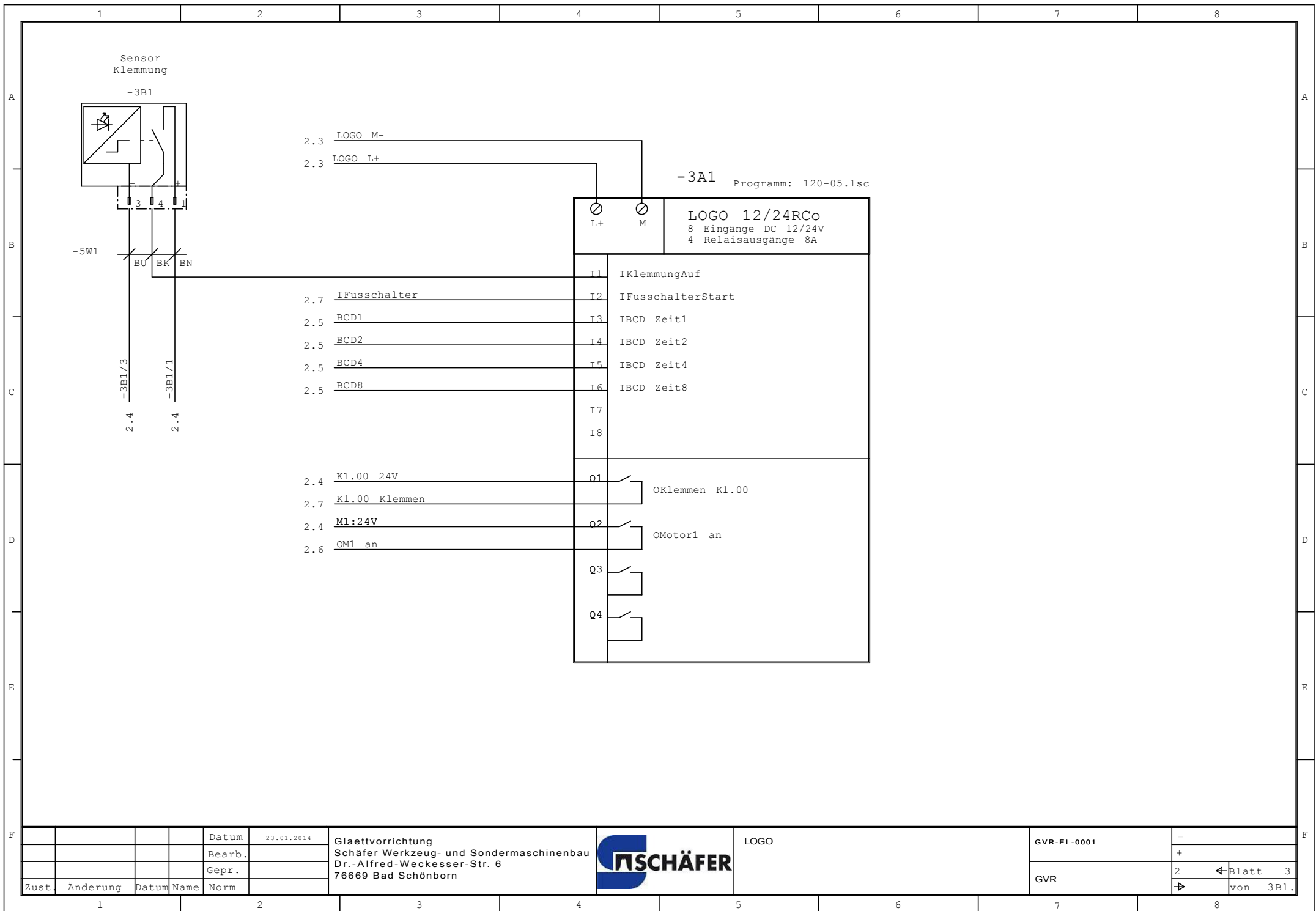


BCD-SCHALTER + MOTOR

GVR-EL-0001

GVR

=	
+	
1	← Blatt 2
→	3 von 3 Bl.



3

2

○

3